

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	22-000465
Award Program	Egg-Tech Phase 2
Project Title	High-throughput in-ovo sexing of chicken eggs using hyperspectral imaging and Raman spectroscopy
Grant Start Date	10/01/2023
Grant End Date	10/01/2024
Total Grant Budget	\$ 499.331,36
Total FFAR Budget	\$ 499.331,36
Matching Funder(s)	n/a
Final Report Date	14/01/2025
Project Director/Principal Investigator Information	
Full Name	Carla van der Pol
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Grantee Organization Information	
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City, State Zip	De Klomp, the Netherlands
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Authorized Signing Official (ASO) Information	

Name	Tjitze Meter
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1. General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Wageningen University & Research – Wageningen, the Netherlands

HatchTech – De Klomp, the Netherlands

Respeggt hatchery – Barneveld, the Netherlands

- 1.2. How many new jobs were created because of this grant funding?

No new jobs were created.

- 1.3. How many jobs were maintained because of this grant funding?

The following jobs were (in part) maintained by the grant during this reporting period through their compensated hours:

- 1 intern (undergraduate student)
- 4 postgraduate researchers
- 3 post-doctoral researchers
- 2 (associate) professors
- 1 staff member at HatchTech
- 2 staff members at Wageningen University

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No.

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No.

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

In commercial poultry production, distinct breeds are used for meat production (broiler type chickens) and for egg production (layer type chickens). The male chicks from layer breeds are culled shortly after hatch because they cannot lay eggs and growing them for meat production is economically unviable. This ethical dilemma may be resolved by sexing the chicken embryos while they are still inside the egg ("in-ovo sexing").

In-ovo sexing is already performed at a commercial scale, but the current procedures all have some disadvantages. Some vision-based techniques only work in a late stage of incubation when the embryo may already be able to experience pain and suffering. Other techniques require a lot of consumables and/or waiting time, resulting in a high cost price or low throughput. In the current project, we use commercially available automation that can extract a small amount of allantoic fluid (the urine of the embryo) from the egg. This technique is used increasingly in the poultry industry. For example, only in Germany, with organic laying hens that have been in-ovo sexed through this system accounts currently up to numbering 14.5 million for organic German laying hens alone. However, throughput speed and use of consumables are factors that can be optimized for this system, improving cost price and acceptability for the poultry industry worldwide.

In the current project, we tried to distinguish male embryos from female embryos, using spectroscopy on the allantoic fluid. Eggs containing a male embryo can then be removed from the incubation process and can be used in animal feed for other purposes, while female embryos are returned to the incubator until they hatch. The allantois harvesting process is harmless, safe, and pain free for the embryos, as it can be performed without touching the embryo and before the onset of brain activity.

Spectroscopic methods involve illuminating the fluid and measuring the wavelength patterns in the reflected or transmitted light. We used both Hyperspectral imaging (HSI) and Raman spectroscopy in order to see whether or not there was a difference between male and female allantoic samples. No difference was found between male and female samples for HSI, and the spontaneous Raman signal in allantoic fluid was found to be absent. Inspired by recent advances in disease diagnosis using surface-enhanced Raman spectroscopy (SERS), the project explored the use of this technique for in-ovo sex determination using allantoic fluid. Using SERS revealed that male and female allantoic fluid samples showed different spectra. Following initial method development, embryonic sex could be predicted with 83% accuracy, but this went down to 70% in larger scale testing. This is not near our desired level of 98% accuracy, which can currently be achieved using PCR analysis (used as the golden standard). However, SERS is a field of research that continues to develop so it may be a promising alternative in the future.

To conclude, the current project did not lead to an innovation that can directly be

implemented for in-ovo sexing by the poultry industry. However, it did show that HSI on raw allantoic fluid is an approach that does not need to be explored further, and that SERS may have some potential in the future.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Currently, allantoic fluid-based analysis methods are being used to sex layer embryos in-ovo. These methods can be used for all layer breeds and strains, as biomarkers or DNA are used to distinguish between the sexes. Although the methods are relatively fast, safe, and affordable, further optimization could be found in making the test quicker, while maintaining accuracy and affordability.

Several alternatives have been studied previously. However, the accuracy of many alternative techniques such as HSI on E7 (Pan et al., 2016), Raman spectroscopy on vitelline blood vessels on E3.5 (Galli et al., 2018), and volatile emissions on E0 (Xiang et al., 2019) did not exceed 91% in experimental setups. This is below the accuracy desired in the poultry industry. Furthermore, their scalability was often poor, throughput time too long, or negative effects were found on hatchability. The current study therefore focused on spectral analysis of allantoic fluid, as this is known to contain relevant biomarkers, and spectroscopy has potential to be quick and low in cost per test.

Unfortunately, the results from the present project are not promising enough to lead to methodology that is directly adoptable by the U.S. agriculture and food systems. HSI results showed that it is unsuitable as an analysis method for raw allantoic fluid. These results can help the U.S. agricultural industry and poultry science in general by showing that this is not a promising route to study further.

Use of Raman spectroscopy, on the other hand, did not show spontaneous signals provide satisfactory results, but while SERS through addition of surface enhancing silver nanoparticles did reveal sex specific differences at 70% accuracy in a larger test set. While this is not close to our desired accuracy of 98%, future developments in the field of SERS may result in better accuracy.

The main benefit of using Raman spectroscopy is that the processing speed can be very high, increasing throughput. In the current PCR method used for in-ovo sexing, time between egg sampling and sexing results is >1 hour. To minimize the effect on embryo temperature and the incubation process, eggs are returned to the incubator after sampling, and are brought out again for sorting into female and male eggs once the results are in. This requires extra labor and movement of the eggs. A fast processing time would allow eggs to be sorted nearly

immediately after they have been sampled.

SERS does require the allantoic fluid sample and added silver nanoparticles to dry before they can be analyzed. Automated equipment would have to be developed to dry the sample quickly, but given that a very small amount is enough, this should be relatively simple to engineer. Another disadvantage is that it does require use of consumables, whereas using raw allantoic fluid for spontaneous Raman signal was expected to be a process without consumables. At an expected \$0,0023 per test, the consumable cost of SERS is very low, however.

In the end, decision making will mostly depend on accuracy. At 70%, the accuracy is not there yet and future developments in the field of SERS will be needed to warrant further research into the method.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Given the limited success of the spectroscopic techniques we studied in the current project, a main outcome is that the agricultural sector should not focus on using HSI on raw allantoic fluid for in-ovo sexing. Similarly, un-enhanced conventional Raman spectroscopy on raw allantoic fluid will does not provide any biologically relevant information and is an area that does not deserve further study for in-ovo sexing, nor for other opportunities such as diagnostics. Although this is disappointing as a result for this project, it does narrow down which research areas are worth exploring for in-ovo sexing.

SERS, on the other hand, does show some potential for future application for in-ovo sexing or in the wider area of diagnostics. As knowledge on SERS progresses and new innovations are presented, this may be worth studying further for its applications in allantoic fluid.

In a wider scope, although HSI did not show promise for raw allantoic fluid, several other applications can be envisioned for the technology. We plan to use the experience and knowledge we gained with the technique within this project in future projects to optimize, for example, blood vessel detection prior to sampling on the Circuit. To obtain clear allantoic fluid

samples that are uncontaminated with blood, blood vessels need to be avoided while lasering and this can be done using vision technology and spectroscopy. Furthermore, embryonic mortality is increased when blood vessels are hit, so avoiding blood vessels makes the process more efficient and optimal. Brown shelled eggs have pigmentation that closely matches the pigments in blood, making blood vessel detection using a simple candling setup difficult. HSI could be used to show a highly contrasted image of developing blood vessels underneath the eggshell regardless of eggshell color when the appropriate wavelength and settings are used.

SERS may also be useful in a wider scope. SERS can be used for PCR readout as an alternative to fluorescence. The experience and knowledge we gained using SERS technology can guide us in the future to study and optimize readout of the current PCR analysis we use for in-ovo sexing within the commercial companies Respeggt and Plantegg.

As an intermediate outcome, results of this study and future projects resulting from it may influence legislation surrounding the timing of in-ovo sexing. It has been shown that the onset of embryonic brain activity as determined through EEG takes place from embryonic day (E)13 onward (Kollmansperger et al., 2023). This emphasizes the appropriateness of using allantoic fluid-based methods for in-ovo sexing, as allantois can be collected without issues from E9 onward, so sexing can take place prior to E12. With the results from this project, allantois fluid-based systems can be further optimized. This may further reduce their cost price and optimize their accuracy, making them competitive alternatives to vision-based methods which operate ideally after E13. The long-term outcomes of this project therefore could strengthen arguments for legislation to set E12 as the latest day on which in-ovo sexing is allowed.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Goal 1: to demonstrate the Circuit's suitability to sample at E8 with minimal loss of hatchability

For Goal 1 (to demonstrate the Circuit's suitability to sample at E8 with minimal loss of hatchability): In January 2024, it was announced that being able to sample at E8 would no longer be a requirement for the Egg-Tech Prize. We therefore did not spend time on this topic within this project from that point onward, but rather focused our time on Goal 2 (tests with SERS). Earlier tests revealed that some modifications to the sampling system Circuit would be necessary to achieve a success rate similar to that of E9.

Goal 2: To demonstrate that chicken embryos can be sexed at >98% accuracy on E8 using Raman spectroscopy and HSI on allantoic fluid.

Milestones:

Q3 2023 – Startup personnell, preparation of lab grade materials

Q4 2023 – Tests on a laboratory scale to create the best setup

Q1 2024 – Improvements in the setup realized; increasing sample size to build the algorithm's robustness with eggs from varying origins

Q2 2024 – Increasing throughput; 10,000 eggs per batch

Each milestone moved up one quarter as the project did not officially start until Q4 of 2023, although one Postdoc had already been hired shortly before the official start of the project in anticipation of starting the experiments.

In Q4 2023, an experiment using HSI was performed to see if there were observable differences in the spectra of male and female embryo allantois samples of brown- and white-shelled eggs. Results were analyzed in Q1 of 2024. It was found that no difference could be observed in spectroscopic signature of male and female samples when using HSI. This approach was therefore put to the side in favor of focusing on Raman spectroscopy.

In Q4 2023, Raman spectroscopy was performed on raw E9 allantois samples in a 384-well plate, using a Raman microscope. No spontaneous Raman signals were observed. These results suggested that Raman signals, if present, would need to be enhanced to be observable in allantois. SERS was tried in Q1 and Q2 of 2024. SERS involves adding nanoparticles, in this case silver, to the raw or processed sample to see if Raman signals appear underneath a Raman microscope. This data was analyzed in Q2 and Q3 of 2024. Results revealed that the most optimal setup resulted in 83% cross-validation accuracy, and 70% accuracy in a larger test set.

Given the limited success of our HSI, Raman, and SERS experiments, the milestone “Improvements in the setup realized; increasing sample size to build the algorithm’s robustness with eggs from varying origins” was completed only partially, and the milestone “Increasing throughput; 10,000 eggs per batch” was not yet started. A more promising spectroscopical solution would have to be found to justify continuing with large batches of eggs to build algorithms.

Goal 3 was to provide evidence that hatcheries are willing to adopt our technology. At the end of 2024, it was announced that Hendrix-ISA LLC (the U.S. distributor of layer breeding company Hendrix Genetics) will adopt the Circuit for in-ovo sexing in Nebraska in early 2025. This will be the first U.S. implemented technology that can sex both white and brown layer strays. The decision to use Respeggt’s Circuit technology shows that the poultry industry in the United States sees merit in the Circuit system and is willing to adopt it.

Furthermore, during the duration of the project, 3 new hatcheries in Europe adopted the Circuit for in-ovo sexing: Steinsland in Bryne, Norway; LOHMANN Deutschland in Ankum, Germany; and Brüterei Weiss in Kirchberg, Germany. The Norwegian hatchery especially shows that our Circuit is a competitive system that provides added value to a hatchery even when there is no national legislation prohibiting chick culling. This hatchery sees the added value of producing chicks without chick culling and getting a return on investment at the retailer.

The German supervisory board KAT supervises traceability of eggs from organic and alternative poultry production systems in Germany and surrounding countries, promoting implementation of their standards throughout Europe. KAT monitors the origin of “free of chick culling” eggs produced within their participating production systems. In 2023, Respeggt

(which uses the Circuit sampling technology) had the largest market share of all in-ovo sexing methods, and had produced 10.9 million laying hens. Plantegg also uses the Circuit for sampling, and they sexed an additional 3.6 million laying hens in 2023. New KAT data has not been publicly released yet, but in March 2024, it was announced that in-ovo sexing is now the main method to avoid male chick culling (over brother fattening) in Germany. Innovative Animal Ag reported that in 2024, 20% of the EU hen flocks have been in-ovo sexed. This shows that in-ovo sexing continues to grow and with the market leaders using the Circuit as sampling technology, it is a viable and profitable option for the poultry industry.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

Goal 1: to demonstrate the Circuit's suitability to sample at E8 with minimal loss of hatchability

No longer an Egg-Tech Prize requirement.

Goal 2: To demonstrate that chicken embryos can be sexed at >98% accuracy on E8 using Raman spectroscopy and HSI on allantoic fluid.

Because being able to sex <E8 is no longer an Egg-Tech Prize requirement, we did not focus on E8.

Within Goal 2, Milestones Q1 2024 – Improvements in the setup realized; increasing sample size to build the algorithm's robustness with eggs from varying origins and Q2 2024 – Increasing throughput; 10,000 eggs per batch were moved forward, because the tried HSI and Raman technology did not yet prove successful enough to continue with larger numbers. Instead, we focused on improving the Raman technique for further experimentation that will likely take place after this project has been wrapped up.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

In Q4 2023, a major objective was to prepare a Circuit for testing on E8 eggs. The Circuit at our Warnder location (the Netherlands) was updated with all available upgrades at that moment.

As a specific objective, we wanted to verify sampling success (number of eggs from which we obtained a clear sample of allantoic fluid of 30 μ l) for our Circuit around embryonic day (E)8, to see whether or not this met our standards or if additional optimization of the Circuit had to take place. In December 2023, a test was performed with Dekalb White eggs from a 63-week-old parent flock that had been stored for 6 days. Eggs were sampled on 4 incubation time points: E7:7H(ours), E7:15H, E8:6H, and E8:12H. Data was collected per tray, per time point, and per robot arm on the Circuit. While we wanted to focus on hatchability, we found that sampling success was lower than desired, and this is an issue we should tackle first. It was found that the best performing robot arm (out of 5 arms) on E8:6H had a missing sample percentage of 9.3%. The robot arm that performed worst showed a missing sample percentage of 19.3% at the same moment, suggesting misalignment of the robot arm, or other technological issues. Normally, percentage of eggs with a missed sample between E9 and E12 is <2%. This led us to conclude that we wanted to identify the cause of missed samples and optimize the Circuit's performance in that area before continuing with hatchability tests. A report of the experiment on E8 sampling can be found in the report "Automated allantoic fluid sampling in embryonated chicken eggs on embryonic day 8".

We did not perform the hatchability tests described in our proposal, because the requirement for E8 sampling is no longer in place. Instead, we spent this time on further spectroscopical experimentation, especially SERS. However, the results on sampling success on E8 are being used outside of this current project to optimize sampling success of the Circuit.

For HSI, objective in the first experiment was to reveal whether or not any sex-specific differences were present in the spectra of raw allantoic fluid samples. Lohmann Brown-Lite and Lohmann LSL-Lite (white) hatching eggs that had been incubated for 8, 9, 10, and 11 days were used, excluding infertile eggs and non-viable embryos. Allantoic fluid samples were collected from 90 eggs/time point. These samples were scanned using a line scan HSI camera as well as through a plate reader. Unfortunately, no sex-specific signature could be found in the allantois. This led to the conclusion that HSI is not a promising spectroscopic technique to use for pure allantoic fluid and it was decided not to continue investigating this approach but focus on Raman spectroscopy for the remainder of the project. A detailed report of the HSI experiments performed within this project can be found in the report "Hyperspectral Imaging of allantoic fluid samples for in-ovo sexing of chicken eggs".

For Raman spectroscopy, the first objective was to study whether or not Raman signals could be observed in raw allantoic fluid. In preliminary tests, we observed that spontaneous Raman signals were absent in the allantois samples. The next objective was therefore to stimulate occurrence of Raman signals by the addition of silver nanoparticles in a technique called SERS. SERS did reveal sex-specific differences in the allantoic fluid samples. Up to 83% cross-validation accuracy was established, which is below the desired goal. A larger experiment later

resulted in 70% accuracy, likely because this dataset had higher variation than the original test set. This means that the goal of 98% accuracy was not accomplished, although we still see some potential in this technique as studies and development in the field of SERS will progress in the future. Protocols, procedures, materials, readout and data analysis of SERS techniques will likely continue to develop and they may be useable for analysis of allantoic fluid.

In the short term, Raman spectroscopy may not be the most promising approach for in-ovo sexing of allantoic fluid. A detailed report of the Raman spectroscopy and SERS experiments performed within this project can be found in the report "Raman spectroscopy and SERS in allantoic fluid samples for in-ovo sexing of chicken eggs".

To summarize, the stated goal "Goal 2: To demonstrate that chicken embryos can be sexed at >98% accuracy on E8 using Raman spectroscopy and HSI on allantoic fluid" was sadly not accomplished for either technique, but some potential was seen for Raman spectroscopy when using SERS technology that could be worth exploring more later.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

For the experiments involving spectroscopy, measurements took place at Wageningen University & Research, under the supervision of Prof. Bas Kemp and Dr. Henry van den Brand. All experiments were performed by trained animal scientists and molecular chemists.

The setup of all experiments followed the scientific principles. All eggs were incubated the same way following protocols set in advance with only the experimental variables changed between treatments (such as incubation day). Spectroscopic measurements were performed blinded, with no sexing results linked to spectroscopy until the point of data analysis. Appropriate statistical analysis was performed on all data to reveal significant differences.

- 2.8. List key stakeholders that could be served by results of this project.

Main providers of in-ovo sampling solutions that can extract allantoic fluid from the egg can use the results of this project to decide on (potentially) alternative methods to analyze the fluid for sexing. Key stakeholders that use the Circuit system for allantoic fluid sampling (which is what the current findings are based on) are:

- Respeggt
- Plantegg

Ultimately, these methods serve layer industry in more particular:

- Hatcheries, that will adopt the technology
- Egg packing stations, which can distribute the eggs at a premium
- Retailers, that can retail the eggs to consumers

However, the poultry industry as a whole may benefit as it can allow separate rearing of males and female broiler chickens:

- Poultry farmers, who can do separate sex rearing for more efficient use of resources
- Slaughterhouses for meat type chickens (broilers), who can benefit from increased uniformity when processing males and females separately
- Retailers, who may develop niche or specialty products from for example extra-large male broilers

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report.

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

HSI unfortunately did not show any sex-specific differences when used on allantoic fluid. Because of the low potential of this approach, we decided to focus our efforts on Raman spectroscopy.

Initial observations of allantoic fluid using Raman spectroscopy revealed that spontaneous Raman signals from allantois were absent. Enhancement would be needed to be able to use this technology for in-ovo sexing. We decided to use SERS to increase the signals.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to

500-word limit)

One colleague attended the Optica Biophotonics Congress: Biomedical Optics in April in Fort Lauderdale (USA). Aim of this congress was to network with specialists in the field of spectroscopy and learn about biomedical optics from global researchers and scientists.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

1 undergraduate student, 4 post-graduate researchers, and 3 post-doctoral researchers worked on this project.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

- Scientific report "Hyperspectral Imaging of allantoic fluid samples for in-ovo sexing of chicken eggs"
- Scientific report "Raman spectroscopy and SERS in allantoic fluid samples for in-ovo sexing of chicken eggs"
- Scientific report "Automated allantoic fluid sampling in embryonated chicken eggs on embryonic day 8"

- 3.2. Please provide a list of citations for the information products produced during the project.

- HatchTech B.V. and Wageningen University & Research, 2024, *Automated allantoic fluid sampling in embryonated chicken eggs on embryonic day 8*, unpublished
- HatchTech B.V. and Wageningen University & Research, 2024, *Hyperspectral Imaging of allantoic fluid samples for in-ovo sexing of chicken eggs*, unpublished
- HatchTech B.V. and Wageningen University & Research, 2024, *Raman spectroscopy and SERS in allantoic fluid samples for in-ovo sexing of chicken eggs*, unpublished

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

No.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

No.

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

No.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

We would prefer it if the report on sampling success on E8 ("Automated allantoic fluid sampling in embryonated chicken eggs on embryonic day 8") and its corresponding dataset (Excel file "Dataset Automated allantoic fluid sampling") remains confidential until 01-10-2026 as this is commercially sensitive information that can be used by competitors to discredit the

Circuit's ability to sample allantoic fluid reliably on incubation days prior to E9. While we currently only focus on sampling after E9 and before E12, we would like to optimize the system for future use on E8 or even earlier, as this will allow more flexibility in hatchery planning. We expect to introduce major revisions on the Circuit in the next year that will greatly improve sampling success <E9.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

At the earlier stages of the project, we did not yet feel that our results were promising enough to present them to a larger audience. The latest SERS results look slightly more promising in terms of accuracy although more work is required to further optimize it and reveal the full potential. We do plan to present this work at a poultry conference in 2025, whether it proves to be successful or not, to share with the poultry industry if this is an approach to study further.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Hyperspectral imaging/VNIR: Excel spreadsheet containing pre-processed exploratory dataset (n=274) containing aggregated HSI reflectance spectra and metadata corresponding to each of the samples.

SERS: Excel spreadsheet containing raw dataset for method development, exploratory experiment and large-scale experiment.

Automated allantoic fluid sampling in embryonated chicken eggs on embryonic day 8: Excel spreadsheet containing raw and processed dataset from sampling on various incubation days.

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

The data sets listed are not related, as they were obtained for different samples of allantoic fluid and/or eggs.



- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.



Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Carla van der Pol

PI Signature: [Signature] Date: 14/01/2025

Authorized Signing Official (ASO) Name: Tjitze Meter

ASO Signature: [Signature] Date: 14/01/2025